Quiz 23:-

C23.

(A) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

Divergence test :-

$\lim_{n\to\infty} \frac{1}{n(\ln(n))^2} = 0$ we know nothing

NO - P-series !, NO Geometric !

Integral test :-

① is $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$ positive on $[2,\infty)$?

yes, a_n is positive for all "n" on $[2,\infty)$.

② is $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$ continuous on $[2,\infty)$?

Let $f(x) = \frac{1}{n(\ln(n))^2}$, $f'(x) = \frac{n(\ln(n))^2(0) - [n(2\ln(n)(\frac{1}{n}) + (\ln n)^2}{(n(\ln(n))^2)^2}$

$\therefore f'(x) = 0$ for critical numbers

$f'(x) = \frac{-2\ln(n) - (\ln(n))^2}{(n^2(\ln(n))^4)} = 0$ when $n = 1$ not on $[2,\infty)$.

(3) is $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$ decreasing on $[2,\infty)$?

$$f'(x) = -\left(\frac{2\ln(n) + (\ln(n))^2}{n^2(\ln(n))^4}\right)$$

$\therefore$ $f'(x)$ is negative which means series is decreasing.

We can use Integral test :-

Let $\int_2^{\infty} \frac{1}{x(\ln(x))^2}\,dx \Rightarrow$ $u = \ln(x)$, $du = \frac{1}{x}\,dx$

$$\Rightarrow \int \frac{du}{u^2} = \int u^{-2}\,du$$

$$= \left[-\frac{1}{u}\right] = \left[-\frac{1}{\ln(x)}\right]$$

$$\Rightarrow \lim_{b\to\infty} \left[-\frac{1}{\ln(x)}\right]_2^b = \lim_{b\to\infty} \left[-\frac{1}{\ln(b)} + \frac{1}{\ln(2)}\right]$$

$$= 1/\ln(2) \neq 0 \neq \infty$$

$\therefore$ $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$ is converges by Integral test.

(B) $S_{\infty} = \frac{1}{2(\ln(2))^2}$

$S_{46} \approx 0.001\,1483$

$S_{47} \approx 0.001\,1358$

$\therefore$ we would need to add 47 terms to get sum with in 0.001

24. (A) $\sum_{n=1}^{\infty} \frac{\sqrt{n}+1}{n^2+3}$

Comparison test:-

Compare to $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2}$

which is $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ P-series $P = \frac{3}{2} > 1$

Convergence by p-series

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}+1}{n^2+3} < \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2}$$

$\therefore \sum_{n=1}^{\infty} \frac{\sqrt{n}+1}{n^2+3}$ is converge by Comparison-

(b) $\sum_{n=0}^{\infty} \frac{5^n}{7^n+n+1}$

Comparison test :-

compare to $\sum_{n=0}^{\infty} \frac{5^n}{7^n} = \sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n$

Converge by Geometric

$|r| = 5/7 < 1$

$$\sum_{n=0}^{\infty} \frac{5^n}{7^n+n+1} < \sum_{n=0}^{\infty} \frac{5^n}{7^n}$$

$\therefore \sum_{n=0}^{\infty} \frac{5^n}{7^n+n+1}$ is converges by comparison.

(c) $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^3}$

Comparison test :-

Compare to $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^3}$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^{5/2}}$ $\Rightarrow$ Converges by p-Series $p = 5/2 > 1$

$$\therefore \sum_{n=1}^{\infty} \frac{\ln(n)}{n^3} < \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^3}$$

$\therefore \sum_{n=1}^{\infty} \frac{\ln(n)}{n^3}$ is converges by a Comparison test.

(25)

(A) $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln(n)}{n^2}$

↓

Alternating we can use A.S.T :-

with $a_n = \frac{\ln(n)}{n^2}$

(1) is positive on $[2,\infty)$? yes

because $\frac{\text{positive}}{\text{positive}}$ = positive for all "n" on $[2,\infty)$

(2) is decreasing on $[2,\infty)$?

$\frac{\ln(n+1)}{(n+1)^2} < \frac{\ln(n)}{n^2}$ yes it is decreasing

② $\lim_{n\to\infty} a_n = 0$?

$\lim_{n\to\infty} \frac{\ln(n)}{n^2} = 0$ because $\left[\ln(n) < n^2\right]$

$\therefore \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln(n)}{n^2}$ is converges by A.S.T (Alternating Series test)

③ $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!}$

↓

Alternating we can use A.S.T with $a_n = \frac{1}{(2n+1)!}$

① is a_n is positive on $[0,\infty)$? yes it is positive for all "n" on $[0,\infty)$ ∴

② is decreasing on $[0,\infty)$? yes because of $a_{n+1} < a_n$

$\Rightarrow \quad \frac{1}{(2n+3)!} < \frac{1}{(2n+1)!}$ ✓

yes.

(3) $\lim_{n\to\infty} \frac{1}{(2n+1)!} = 0$ yes by using limit.

$\therefore \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!}$ is converge by A.S.T.